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Case 7 — Ms. Anika Gupta | 41 F · Slow-onset Cerebellar Ataxia · Unknown Aetiology

Target: 70%+ | Neurological Physiotherapy

■ PROBLEM LIST

- Dysmetria — limb incoordination on reaching tasks
- Dysdiadochokinesia — impaired rapid alternating movements
- Truncal and gait ataxia — wide-based, unsteady gait
- Cerebellar dysarthria — slurred, scanning speech
- Impaired postural stability and dynamic balance
- Fine motor dysfunction — grip and object manipulation
- Unable to safely walk on uneven terrain or in crowds
- Difficulty using cutlery — nutrition and independence at risk
- Cannot fasten buttons or do up zips — reduced ADL independence
- Reduced community mobility and social participation
- Potential impact on employment depending on role demands

■ RED FLAGS

None identified

■ PATHOPHYSIOLOGY

• The key output neurons are Purkinje cells , which provide the cerebellum's sole inhibitory output to the deep nuclei.
• In cerebellar ataxia, Purkinje cells are progressively damaged or lost — without their inhibitory influence, motor output from the deep nuclei becomes disorganised, irregular in timing, and inaccurate in force scaling.
• The cerebellar vermis and spinocerebellar tracts govern axial and gait coordination — degeneration here produces the wide-based, lurching "drunken" gait.
• Cerebellar connections to the brainstem also coordinate articulatory muscles, so Purkinje cell loss produces the characteristic scanning dysarthria heard in Anika's speech.

■ DIFFERENTIAL DIAGNOSES

1. Supporting features:
- Supporting features:

2. Supporting features:
- Supporting features:
- Progressive bilateral cerebellar syndrome in an adult. However, alcoholic ataxia preferentially affects the
- anterior and superior vermis

3. Supporting features:
- Supporting features:

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Finger-to-Nose Test (FNT)
MEASURES Limb dysmetria and intention tremor — integrity of cerebellar hemisphere coordination and sensorimotor calibration HOW TO PERFORM Seat the patient comfortably. Position yourself opposite, holding your index finger at arm's length from the patient. Ask the patient to touc

Heel-to-Shin Test (HTS)
MEASURES Lower limb cerebellar coordination — tests cerebellar hemisphere and spinocerebellar tract function in the legs HOW TO PERFORM Ask the patient to lie supine on the plinth. Instruct them to place the heel of one foot precisely on the opposite knee. Ask them to slide the h

Romberg Test (Eyes Open vs. Closed)
MEASURES Distinguishes cerebellar ataxia (unstable with eyes OPEN) from sensory/dorsal column ataxia (unstable only with eyes CLOSED) — identifies which sensory modality is most contributing to instability HOW TO PERFORM Ask the patient to stand with feet together, arms at sides

■ TREATMENT PLAN

■ Initial Phase — Weeks 1–4 (2–3 sessions/week)
- Neurophysio Rationale
- Cerebellar rehabilitation leverages
- error-based motor learning
- neural plasticity
- Exercise 1 — Seated Target Reaching (Upper Limb Coordination)

■ Long-Term Plan — Weeks 4–12+
- ■ Examiner Demonstration — Techniques Performable in the Room

• Guided Reaching Task:
• Tandem Stance Challenge: